

# Improved Fusion Algorithm for Fiber-Optic Gyroscope: Suppressing Intrinsic Noise and Environmental Disturbances

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**Abstract**—In the field of precision measurements, highly sensitive interferometric fiber-optic gyroscopes (IFOGs) hold significant value. However, the application of IFOGs is constrained by intrinsic noise and environmental disturbances. This study conducts a comprehensive noise analysis based on the conventional phase generated carrier (PGC) algorithm. On the basis of noise theory, an improved fusion algorithm is proposed in conjunction with a light source sharing configuration to suppress the primary intrinsic noise—relative intensity noise (RIN) and reduce environmental disturbances. Experiments under excitation by piezoelectric transducers (PZTs) and environmental disturbances are carried out to verify the effectiveness of the scheme. In comparison to the minimum configuration using the conventional PGC algorithm, the new scheme achieves a 16.5-fold suppression in self-noise, 16.0-fold suppression in angular random walk, and 10.0-fold suppression in bias instability. More importantly, the ability to suppress environmental disturbances is demonstrated. The proposed scheme demonstrates the superior ability to make high-precision measurements in complex environments, showcasing world-leading performance and offering insights into the application of IFOGs in a broader range of measurement scenarios.

**Index Terms**—Interferometric fiber-optic gyroscope, improved fusion algorithm, intrinsic noise suppression, environmental disturbance suppression.

## I. INTRODUCTION

THE interferometric fiber optic gyroscope (IFOG) serves as a sensor for measuring rotational motion [1]. Its all-solid-state design, compact structure and high stability [2], [3] have contributed to its widespread use in various applications, including inertial navigation [4], [5], rotational seismic wave measurements [6], gravity gradient measurements [7] and universal time measurements [8]. However, intrinsic noises and

environmental disturbances pose challenging issues for IFOGs, leading to sensitivity degradation and restricting their application scenarios.

The intrinsic noises encompass relative intensity noise (RIN), thermal phase noise (TPN), shot noise, and Johnson-Nyquist noise [1], [9]. RIN is the primary intrinsic noise component, originating from the beat effect among uncorrelated frequency components in a broadband light source [10], [11]. This is reflected as random fluctuations in the optical power of the source. Additionally, environmental disturbances, such as vibration, present significant challenges for IFOGs [12]. To suppress RIN and environmental disturbances, modulation depth control, optical configuration improvements, and data processing approaches have emerged. RIN suppression can be achieved when the modulation depths for sinusoidal and square-wave modulations are around 2.7 [13] and  $7/8\pi$  rad [14], respectively. However, the effectiveness of this approach is limited, while changes in modulation depth often influence other noises such as shot noise [9]. Optical structure improvements typically involve adding an optical reference arm, which entails extracting a separate path from the light source section to detect optical power fluctuations [15], [16], [17]. However, the inevitable inconsistency between the reference and measured signals limits the effectiveness of RIN suppression [17], [18]. Data processing methods such as empirical mode decomposition [19], [20] and artificial neural networks [21], [22] have been employed to address environmental disturbances. Nevertheless, these algorithms often demand high-quality training data and may lack generalization ability when confronted with more diverse environmental conditions.

In recent years, fusion of multi-sensor devices for intrinsic noise and environmental disturbance suppression has been prominent in various fields [23], [24], [25], [26]. Inspired by this idea, fusion in IFOGs is proposed to achieve noise and disturbance suppression [12]. In the study, environmental disturbances among adjacent sensors exhibit similarities, and the fusion scheme implemented through common-mode and differential-mode measurements proves effective in suppressing disturbances. The scheme is both simple and effective. It is valuable to carry out further research along with this study. However, the conventional phase generated carrier (PGC) demodulation is employed for measurement, and as a result, the intrinsic noise can not be handled well, the residual noise

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deteriorates the sensitivity of the configuration, and limits the further development of the IFOG in weak signal detection. Designing a new demodulation algorithm that achieves complete noise suppression while effectively removing environmental disturbances is of great importance.

In this study, an improved fusion algorithm in concert with a light source sharing configuration is studied and validated. The proposed scheme combines the inherent high stability of PGC [27], [28] with the capability to suppress intrinsic noises and environmental disturbances. It is structurally simple and efficient, avoiding complex optical path modification and the lack of generalization ability caused by various algorithms. By adjusting the phase difference of the two IFOGs, it is possible to flexibly select the detection modes of common-mode and differential-mode. The improved fusion algorithm jointly processes the harmonic signals in two sensors, separates the common-mode signals and the differential-mode signals, and selectively extracts the required components. More importantly, in the process of joint data processing, the influence of RIN on harmonics is suppressed. In order to verify the validity of the theory, a series of verification experiments are carried out. We use two sets of piezoelectric transducers (PZTs) to apply common-mode and differential-mode excitation to two IFOGs, respectively. The improved fusion algorithm provides excellent signal extraction and noise suppression in each measurement mode. Furthermore, the system is tested in an environment with complex disturbances and also shows excellent performance. After cooperative data processing, the proposed scheme achieves a remarkable suppression, with the root power spectral density (rPSD), angular random walk (ARW), and bias instability (BI) reduced by an impressive factor of 16.5, 16.0, and 10.0 times, respectively, outperforming the performance of individual IFOGs.

This article is organized as follows. In Section II, we provide information on the sensor setup, focusing on the source sharing configuration and modulation scheme. In Section III, the conventional PGC algorithm is introduced. Furthermore, we discuss the impact of four intrinsic noises and point out that RIN dominates as the primary intrinsic noise. In Section IV, we introduce an improved fusion algorithm. Theoretical analysis demonstrates its ability to eliminate differential-mode disturbances during common-mode measurements and effectively suppress RIN-induced noise power in harmonic extraction. Similarly, in the differential-mode measurement, it efficiently suppresses both common-mode disturbances and RIN influence. In Section V, we design and conduct two experiments: sinusoidal excitation extraction and disturbance removal. The effectiveness of the proposed scheme for the suppression of environmental disturbances and RIN is verified.

## II. SENSOR SETUP

The Sagnac effect refers to the phase shift experienced by light traveling in opposite directions within a rotating optical system. The accumulated Sagnac phase within the optical fiber coil is related to the rotation rate vector  $\Omega$  and the equivalent area vector  $\mathbf{A}$ . Its expression is given by (1) [1].

$$\phi_s = \frac{4w}{c^2} \mathbf{A} \cdot \Omega \quad (1)$$

In (1),  $w$  is the central frequency of light source,  $c$  is the speed of light in a vacuum.  $\mathbf{A}$  is defined in terms of the line integral, expressed as  $\mathbf{A} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{r}$ , where  $\mathbf{r}$  is radial coordinate vector. The Sagnac effect appears as the flux of the rotation vector through the closed path of the equivalent area vector, which implies that the sensitive direction of fiber coil impacts the sign of the Sagnac phase.

The sensor setup is depicted in Fig. 1. The optical propagation process begins with an amplified spontaneous emission (ASE) light source. Following passage through a polarizer, the light becomes linear polarized and is subsequently divided into two equal-power beams via a 50/50 coupler. The shared light source, coupled with the preceding polarizer, guarantees a high correlation of intrinsic noise between the two IFOGs. To split and modulate the linearly polarized light, two multifunctional integrated optical chips (MIOC) are employed. Two sinusoidal signals of the same frequency  $f_h$  are applied to MIOC A and MIOC B. Both fiber coils share the same length  $L$  and diameter  $D$ . Dynamic phase modulation is implemented to suppress low-frequency noise, such as temperature variations [29]. The intrinsic modulation frequency of the fiber coil is denoted as  $f_0 = 1/2\tau$ , where  $\tau$  represents the light transit time and is characterized as  $\tau = n_{eff}L/c$ , with  $n_{eff}$  being the effective refractive index. The modulation frequency  $f_h$  is typically chosen as an odd multiple of the intrinsic frequency  $f_0$  [1], [29]. The modulation depth of both IFOGs is set to  $\phi_b$ . We employ a reference signal as the trigger clock for two signal generators to precisely control modulation phase difference  $\Delta\theta$ . The back-propagating optical signals carrying the Sagnac phases enter the photodetectors (PDs) through the circulators. All the optical components possess polarization-maintaining capabilities. PD A and PD B respectively convert the optical interference signals into electrical signals, which are then subjected to analog-to-digital conversion and recorded by the data acquisition system.

In Fig. 1, two polarization-maintaining fiber (PMF) coils are placed close to each other and sensitive directions of the both are coaxial. The Sagnac phases within the two IFOGs are denoted as  $\phi_{sA}$  and  $\phi_{sB}$  and based on the configuration principle, it holds that  $|\phi_{sA}| = |\phi_{sB}| = |\phi_s|$ . In practice, both outputs of IFOGs contain environmental disturbances, denoted as  $N_A$  and  $N_B$ . Mathematically, disturbances can be divided into common-mode disturbance  $N_+$  and differential-mode disturbance  $N_-$ . The output phases are expressed as [12]:

$$\phi_A = \phi_{sA} + N_+ + N_-, \quad (2a)$$

$$\phi_B = \phi_{sB} + N_+ - N_-. \quad (2b)$$

Fortunately, there is often a dominant type of environmental disturbance. Flexibly choosing the measurement mode can effectively suppress a significant amount of environmental disturbances [12]. In the presence of dominant differential-mode noise, the sensitive directions of the two fiber coils are aligned in the same direction, that is  $\phi_{sA} = \phi_{sB} = \phi_s$ . Subsequently, the common-mode measurement is carried out. The output of the fusion scheme is  $\phi_c \equiv (\phi_A + \phi_B)/2 = \phi_s + N_+$ . In the presence of dominant common-mode noise, the sensitive directions of the two fiber coils are aligned in the opposite direction, that is  $\phi_{sA} = -\phi_{sB} = \phi_s$ . The differential-mode measurement

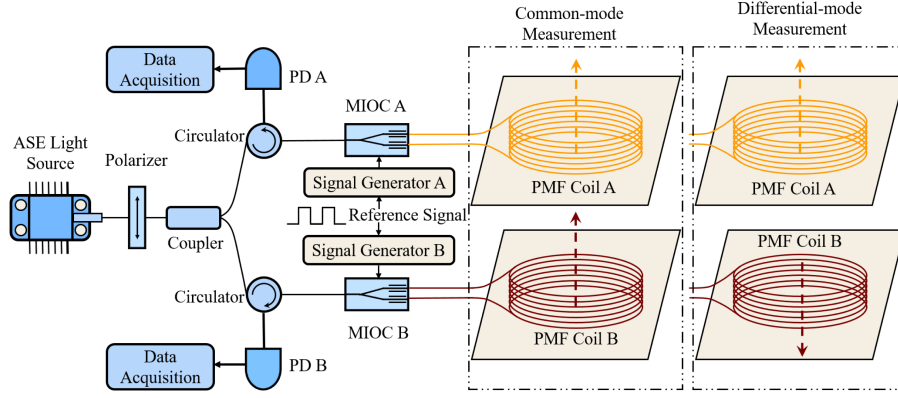


Fig. 1. Sensor setup: Two IFOGs share the same ASE light source. The fusion scheme includes common-mode and differential-mode measurement. Two fiber coils are placed close to each other and sensitive directions of the both are coaxial. Two signal generators share a single reference signal.

is then carried out. The output of the fusion scheme is  $\phi_d \equiv (\phi_A - \phi_B)/2 = \phi_s + N_-$ .

The fusion scheme can be applied in various fields. For general applications such as inertial navigation systems and rotational seismology, we can configure the sensitive directions of the two IFOGs in the same orientation to jointly measure the rotational motion of the inertial body. The fusion scheme can use common-mode measurement to suppress the differential-mode disturbances in the environment. For gradient field measurement, such as gravity gradient, we can configure two adjacent IFOGs with their sensitive directions oriented oppositely. This suppresses a significant amount of common-mode noise in the environment, such as vibrations [7], [12].

### III. PGC DEMODULATION AND NOISE ANALYSIS

As mentioned in Section II, dynamic sinusoidal modulations are used to mitigate the influence of low-frequency noise on the Sagnac phase. High-precision IFOGs commonly employ the PGC algorithm for signal demodulation. PGC offers a method for extracting phase information from modulated signals, enhancing the robustness of phase information extraction and improving measurement precision in IFOGs [30], [31], [32]. The phase demodulation process is illustrated below. The conventional PGC algorithm is designed to solve angular velocity from IFOG A and B independently, so the phase difference  $\Delta\theta$  has been ignored. The optical signals detected by PDs can be expressed as (3) through the Bessel expansion.

$$\begin{aligned}
 I_i &= I_0 [1 + \cos(\phi_i + \phi_b \cos 2\pi f_h t)] \\
 &= I_0 \{1 + J_0(\phi_b) \cos \phi_i \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(\phi_b) \cos(4n\pi f_h t) \cos \phi_i \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(\phi_b) \cos[(4n-2)\pi f_h t] \sin \phi_i\}
 \end{aligned} \quad (3)$$

In (3),  $I_0$  is the average light intensity,  $t$  represents time, the index  $i$  takes values A or B, and  $J$  represents the Bessel function. The

amplitudes of the first and second harmonic components  $S_{i1}$ ,  $S_{i2}$  are extracted from the Fourier transform of  $I_i$ . Subsequently, a division operation in concert with an arctangent operation is performed using  $S_{i1}$  and  $S_{i2}$  to calculate  $\phi_i$  [30]. The expression is shown as follows.

$$\begin{aligned}
 S_{i1} &= -2I_0 J_1(\phi_b) \sin \phi_i, \\
 S_{i2} &= -2I_0 J_2(\phi_b) \cos \phi_i.
 \end{aligned} \quad (4)$$

$$\phi_i = \arctan \left[ \frac{S_{i1} J_2(\phi_b)}{S_{i2} J_1(\phi_b)} \right]. \quad (5)$$

During the process of calculating the Sagnac phase, numerous intrinsic noises can result in a degradation of sensitivity. The intrinsic noises consist of RIN, TPN, shot noise, and Johnson-Nyquist noise.

To handle the parasitic interference caused by Rayleigh backscattering, broadband light sources are widely employed due to their shorter coherence lengths. However, random beats between uncorrelated frequency components in broadband sources lead to fluctuations in the output optical power [9], [13]. The random fluctuation noise referred to as RIN. In the optical configuration illustrated in Fig. 1, an ASE light source is employed. Therefore, the light intensity correlation function  $\Gamma_I(\tau)$  is expressed as the fourth-order correlation function of the electric field  $E(t)$  [33]. Besides, the transmitted light waves remain linearly polarized because all components are polarization-maintaining. Thus,  $\Gamma_I(\tau)$  is further expressed as the second-order correlation of the intensity function  $I$  [33]:

$$\begin{aligned}
 \Gamma_I(\tau) &= \langle E(t)E^*(t)E(t+\tau)E^*(t+\tau) \rangle \\
 &= I^2 [1 + |\gamma(\tau)|^2],
 \end{aligned} \quad (6)$$

where  $\langle \rangle$  denotes expectation operator and  $\gamma(\tau)$  represents the coherence degree of light. The PSD of RIN, denoted as  $S_{RIN}(f)$ , is the Fourier transform of the correlation function  $\Gamma_I(\tau)$ .

$$S_{RIN}(f) = \mathcal{F}[\Gamma_I(\tau)] = I^2 \int_0^{+\infty} p(v)p(v+f)dv, \quad (7)$$

where  $p(v)$  denotes the normalized PSD. The frequency bandwidth of ASE source is approximately 2 THz. In contrast, both



the photodetectors and analog-to-digital converters have much narrower bandwidths, typically within the range of a few megahertz. Therefore, in the case of an IFOG utilizing a rectangular spectrum ASE light source,  $S_{RIN}(f)$  is demonstrated to be inversely proportional to the spectral width  $\Delta f_s$ .

$$S_{RIN}(f) = S_{RIN}(0) = I^2 \int_0^{+\infty} p^2(v)dv = \frac{I^2}{\Delta f_s}. \quad (8)$$

Thermal phase noise (TPN) is a fundamental noise source in optical fiber interferometers, arising from thermal fluctuations in the refractive index of optical fibers. It manifests as phase noise superimposed on the Sagnac phase shift. In the demodulated angular velocity, the noise power spectrum  $S_{TPN}(f)$  caused by TPN is expressed by (9) [7].

$$S_{TPN}(f) = \frac{\lambda c}{2\pi L D J_1^2(\phi_b)} \sum_{n=0}^{\infty} J_{2n+1}^2(\phi_b) \{ \psi^2[(2n+2)f_h] + \psi^2(-2nf_h) \}^{1/2},$$

$$\psi(f) = \sqrt{\frac{k_B T^2 L}{\kappa \lambda^2} \left( \frac{dn_{eff}}{dT} + n_{eff} \alpha \right)^2}$$

$$\sqrt{\ln \left[ \frac{\left( \frac{2}{W_0} \right)^4 + \left( \frac{2\pi f}{\xi} \right)^2}{\left( \frac{4.81}{d} \right)^4 + \left( \frac{2\pi f}{\xi} \right)^2} \right] \times \left[ 1 - \text{sinc} \left( \frac{2\pi n_{eff} f L}{c} \right) \right]}. \quad (9)$$

$\lambda$  is the central wavelength,  $k_B$  is the Boltzmann constant, and  $T$  is the absolute temperature in Kelvin.  $\kappa$  is the thermal conductivity.  $L$  is the total length of the fiber coil.  $\xi$  is the thermal conductivity coefficient.  $n_{eff}$  is the effective refractive index,  $dn_{eff}/dT$  is the derivative of the refractive index with respect to temperature, and  $\alpha$  is the linear coefficient of thermal expansion.  $W_0$  and  $d$  are the mode field radius and cladding diameter of the fiber, respectively. The two terms separated by the symbol  $\times$  in (9) characterize the exponential attenuation of thermodynamic waves in the fiber and the delay filtering effect introduced by the interferometer [34], [35], respectively. Because TPN attenuates with increasing frequency, high-order eigenfrequency modulation can effectively suppress the influence of this thermodynamic disturbance [29].

Shot noise is generated from discrete flows of charge carriers in detectors, setting the quantum limit of IFOG. It is modeled as Gaussian white noise superimposed on the detected signal. Johnson-Nyquist noise is generated by the carrier fluctuation within the trans-resistance amplifier. It is a white noise introduced by the detection system and appears as power fluctuations in the dissipative circuit. The power spectrum of noise caused by shot noise  $S_{shot}(f)$  and Johnson-Nyquist noise  $S_{J-N}(f)$  in the demodulated results are expressed by (10) and (11) [7], [36], respectively.

$$S_{shot}(f) = \frac{\lambda c}{2\pi L D} \sqrt{\frac{2e[1 + \cos \phi_b]}{\eta I \sin^2 \phi_b}}. \quad (10)$$

$$S_{J-N}(f) = \frac{\lambda c}{2\pi L D} \sqrt{\frac{4k_B T}{R \eta^2 I^2 \sin^2 \phi_b}}. \quad (11)$$

TABLE I  
SUMMARY OF PHYSICAL PARAMETERS HIGHLY RELATED TO NOISE PERFORMANCE

| Label         | description                                 | Value                                           |
|---------------|---------------------------------------------|-------------------------------------------------|
| $\lambda$     | Central wavelength                          | 1550nm                                          |
| $f_h$         | Modulation frequency                        | 297.9kHz                                        |
| $\kappa$      | Thermal conductivity                        | $1.37 \text{ W m}^{-1} \text{ K}^{-1}$          |
| $T$           | Temperature                                 | 300K                                            |
| $n_{eff}$     | Effective refractive index                  | 1.47                                            |
| $dn_{eff}/dT$ | Temperature coefficient of refractive index | $1.1 \times 10^{-5} \text{ K}^{-1}$             |
| $\alpha$      | Thermal expansion coefficient               | $5.5 \times 10^{-7} \text{ K}^{-1}$             |
| $d$           | Cladding diameter                           | $80 \mu\text{m}$                                |
| $W_0$         | Mode field radius                           | $5.25 \mu\text{m}$                              |
| $\xi$         | Thermal conductivity                        | $8.2 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$ |
| $\eta$        | responsivity of PD                          | 0.9 A/W                                         |
| $R$           | resistor                                    | 1 k $\Omega$                                    |

$e$  is the charge on an electron, and  $\eta$  represents photodetector responsivity.  $R$  denotes the detector resistance. Shot noise is highly correlated with the detected light intensity, and Johnson-Nyquist noise is highly correlated with the detector resistance. Appropriate regulation of modulation depth, light source power, and detector resistance can inhibit both to a certain extent.

To provide a thorough insight into the impact of intrinsic noises, we conduct both numerical simulations and theoretical calculations. All parameters are kept consistent with subsequent experiments. The PMF coil length  $L$  is set to 1 km, and the diameter  $D$  is set to 0.13 m. The Sagnac phase  $\phi_s$  is set to  $8 \times 10^{-5}$  rad, simulating the common situation of measuring the rotation of the Earth. The influence of RIN is simulated by numerical simulation. According to the definition of RIN, the optical intensity  $I_0$  can be decomposed into a non-time-varying constant  $I^{dc}$  and a random fluctuation component  $I^{RIN}$ . The average light intensity  $I^{dc}$  detected by PD is 2 dBm (1.6 mW). RIN is modeled as a Gaussian random process superimposed on  $I^{dc}$  and the level of RIN is determined by the signal-to-noise ratio of  $I^{dc}$ , that is,  $20\log_{10}(I^{dc}/I^{RIN})$ . The parameter is set to 58 dB in the simulation [9]. The effects of TPN, shot noise, and Johnson-Nyquist noise are calculated using (9), (10) and (11). Relevant parameters are summarized in Table I. The modulation depth ranges from 1.5 rad to 3.0 rad in simulation, and the results are illustrated in Fig. 2.

Some conclusions can be drawn from Fig. 2. Across all investigated modulation depths, RIN is the predominant intrinsic noise. When the modulation depth is 2.7 rad, the impact of RIN on demodulation is minimized. The application of high-order frequency modulation allows TPN to reach a level comparable to shot noise. Directly deriving common-mode and differential-mode signals by adding or subtracting the outputs of two IFOGs introduces a considerable amount of intrinsic noise, contaminating the results and submerging faint signals within the noise. Conventional PGC algorithms prove insufficient in mitigating the interference from intrinsic noise. Hence, the development of an efficient method for extracting common-mode and differential-mode signals while concurrently suppressing intrinsic noise holds substantial significance.

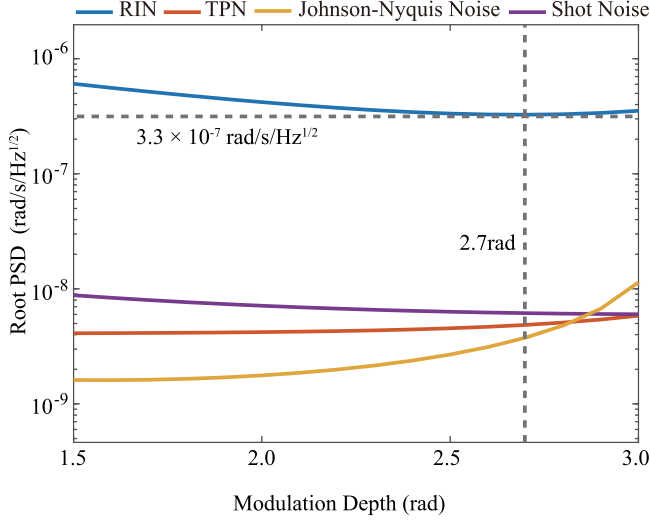


Fig. 2. Noise decomposition under the conventional PGC algorithm. RIN is the primary intrinsic noise, which sets the measurement limit and its impact minimized at a modulation depth of 2.7 rad. Due to the high-order frequency modulation, TPN is effectively suppressed, even less than the shot noise.

#### IV. IMPROVED FUSION ALGORITHM

An improved fusion demodulation algorithm is proposed to achieve disturbance suppression through common-mode and differential-mode measurement while effectively suppressing RIN. Adjusting the phase difference  $\Delta\theta$  between the two modulation signals enables a switch in measurement modes. For common-mode signal extraction, the coil sensitive directions is adjusted the same and  $\Delta\theta$  is adjusted to  $\pi$  rad. Furthermore, the optical intensity signals in PD A and PD B can be expressed as follows. Compared to (3),  $\phi_A$  and  $\phi_B$  are replaced with  $\phi_s + N_+ + N_-$  and  $\phi_s + N_+ - N_-$ , respectively, to facilitate a clearer derivation. Each Bessel expansion consists of three components: the first represents the DC component of the detection signal, the second corresponds to the even harmonic frequencies of the carrier signal, and the third corresponds to the odd harmonic frequencies. It is worth noting that since the modulation phase difference  $\Delta\theta$  is  $\pi$  rad, the signs of the third terms in (12) and (13) are reversed.

$$\begin{aligned}
 I_A &= I_0 [1 + \cos(\phi_s + N_+ + N_- + \phi_b \cos 2\pi f_h t)] \\
 &= I_0 \{1 + J_0(\phi_b) \cos(\phi_s + N_+ + N_-) \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(\phi_b) \cos(4n\pi f_h t) \cos(\phi_s + N_+ + N_-) \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(\phi_b) \cos[(4n-2)\pi f_h t] \\
 &\quad \sin(\phi_s + N_+ + N_-)\}. \quad (12) \\
 I_B &= I_0 [1 + \cos(\phi_s + N_+ - N_- - \phi_b \cos 2\pi f_h t)] \\
 &= I_0 \{1 + J_0(\phi_b) \cos(\phi_s + N_+ - N_-)
 \end{aligned}$$

$$\begin{aligned}
 &+ 2 \sum_{n=1}^{\infty} (-1)^n J_{2n}(\phi_b) \cos(4n\pi f_h t) \cos(\phi_s + N_+ - N_-) \\
 &- 2 \sum_{n=1}^{\infty} (-1)^n J_{2n-1}(\phi_b) \cos[(4n-2)\pi f_h t] \\
 &\sin(\phi_s + N_+ - N_-)\}. \quad (13)
 \end{aligned}$$

By performing subtraction and addition operations using the detected signals, we obtain  $I_{sub}$  and  $I_{add}$  and the expressions are illustrated in (14) and (15). The common-mode and differential-mode components have been separated.

$$\begin{aligned}
 I_{sub} &= \frac{I_A - I_B}{2} \\
 &= -I_0 J_0(\phi_b) \sin N_- \sin(\phi_s + N_+) \\
 &\quad - 2 \sum_{n=1}^{\infty} (-1)^n I_0 J_{2n}(\phi_b) \cos(4n\pi f_h t) \sin N_- \sin(\phi_s + N_+) \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n I_0 J_{2n-1}(\phi_b) \cos[(4n-2)\pi f_h t] \\
 &\quad \cos N_- \sin(\phi_s + N_+). \quad (14)
 \end{aligned}$$

$$\begin{aligned}
 I_{add} &= \frac{I_A + I_B}{2} \\
 &= I_0 + I_0 J_0(\phi_b) \cos N_- \cos(\phi_s + N_+) \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n I_0 J_{2n}(\phi_b) \cos(4n\pi f_h t) \cos N_- \cos(\phi_s + N_+) \\
 &\quad + 2 \sum_{n=1}^{\infty} (-1)^n I_0 J_{2n-1}(\phi_b) \cos[(4n-2)\pi f_h t] \\
 &\quad \sin N_- \cos(\phi_s + N_+). \quad (15)
 \end{aligned}$$

The amplitudes of the first and the second harmonic components are extracted from the third component of  $I_{sub}$  and the second component of  $I_{add}$ , respectively. The extracted harmonic amplitudes are illustrated in (16). Compared to the conventional PGC demodulation algorithm in (4), differential-mode disturbance  $N_-$  is separated from the Sagnac phase  $\phi_s$ . By employing the method of calculating the harmonic ratio in the PGC algorithm, differential-mode disturbance  $N_-$  can be eliminated from the results, as expressed in (17).

$$\begin{aligned}
 S_1^{cm} &= -2I_0 J_1(\phi_b) \sin(\phi_s + N_+) \cos N_-, \\
 S_2^{cm} &= -2I_0 J_2(\phi_b) \cos(\phi_s + N_+) \cos N_-. \quad (16)
 \end{aligned}$$

$$\phi_c = \phi_s + N_+ = \arctan \left[ \frac{S_1^{cm} J_2(\phi_b)}{S_2^{cm} J_1(\phi_b)} \right]. \quad (17)$$

In the differential-mode measurement, we reverse the sensitive direction of coil B and set  $\Delta\theta$  to 0.  $\phi_A$  and  $\phi_B$  are replaced with  $\phi_s + N_+ + N_-$  and  $-\phi_s + N_+ - N_-$ , respectively. Similar to the derivation process for common-mode measurement, we continue to extract the amplitude of the first harmonic component from  $I_{sub}$  and the amplitude of the second harmonic component

from  $I_{add}$ . The extracted amplitudes are presented in (18). Compared to the conventional PGC algorithm, the harmonic amplitudes are now multiplied by a factor of  $\cos N_+$ . Subsequently, the differential-mode phase  $\phi_s + N_-$  is calculated from the ratio of  $S_1^{dm}$  and  $S_2^{dm}$ .

$$\begin{aligned} S_1^{dm} &= -2I_0 J_1(\phi_b) \sin(\phi_s + N_-) \cos N_+, \\ S_2^{dm} &= -2I_0 J_2(\phi_b) \cos(\phi_s + N_-) \cos N_+. \end{aligned} \quad (18)$$

$$\phi_d = \phi_s + N_- = \arctan \left[ \frac{S_1^{dm} J_2(\phi_b)}{S_2^{dm} J_1(\phi_b)} \right]. \quad (19)$$

In such a fusion scheme to suppress environmental disturbance, effective RIN suppression is concurrently achieved. According to the noise model, the light intensity  $I_0$  can be decomposed into the DC term  $I^{dc}$  and the random fluctuation term  $I^{RIN}$ . The DC term  $I^{dc}$  contributes to harmonic signals, while the random fluctuation term  $I^{RIN}$  introduces noise in the demodulation process.

To demonstrate the efficacy of this scheme in RIN suppression, we first compare the noise power caused by RIN in  $I_A$ ,  $I_{sub}$ , and  $I_{add}$  during common-mode measurement. The noise power for each is denoted as  $P_{PD}$ ,  $P_{sub}^{cm}$ , and  $P_{add}^{cm}$ , respectively. The calculation of noise powers takes advantage of the orthogonality between different frequency sinusoidal components and the statistical independence of common-mode and differential-mode phases. Additionally, given the extreme weakness of the phases, the sine and cosine terms in the equations are replaced by their first-order approximations through Taylor expansion.

$$\begin{aligned} P_{PD} &= I^{RIN^2} [1 + J_0^2(\phi_b) + 2J_0(\phi_b) \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \langle \cos^2(\phi_s + N_+ + N_-) \rangle \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) \langle \sin^2(\phi_s + N_+ + N_-) \rangle] \\ &= I^{RIN^2} [1 + J_0^2(\phi_b) + 2J_0(\phi_b) + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) (P_s + P_+ + P_-)]. \end{aligned} \quad (20)$$

$$\begin{aligned} P_{sub}^{cm} &= I^{RIN^2} [J_0^2(\phi_b) \langle \sin^2(N_-) \sin^2(\phi_s + N_+) \rangle \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \langle \sin^2(\phi_s + N_+) \sin^2 N_- \rangle \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) \langle \sin^2(\phi_s + N_+) \cos^2 N_- \rangle] \\ &= I^{RIN^2} [J_0^2(\phi_b) (P_s + P_+) P_- \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) (P_s + P_+) P_- \end{aligned}$$

$$+ 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) (P_s + P_+)]. \quad (21)$$

$$\begin{aligned} P_{add}^{cm} &= I^{RIN^2} [1 + J_0^2(\phi_b) + 2J_0(\phi_b) \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \langle \cos^2(\phi_s + N_+) \cos^2 N_- \rangle \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) \langle \cos^2(\phi_s + N_+) \sin^2 N_- \rangle] \\ &= I^{RIN^2} [1 + J_0^2(\phi_b) + 2J_0(\phi_b) + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) P_-]. \end{aligned} \quad (22)$$

In (20), (21) and (22),  $P_s$ ,  $P_+$  and  $P_-$  denote the phase power. Specifically,  $P_s = \langle \phi_s^2 \rangle$ ,  $P_+ = \langle N_+^2 \rangle$  and  $P_- = \langle N_-^2 \rangle$ . In weak signal detection,  $P_s$ ,  $P_+$  and  $P_-$  are extremely weak. In contrast to the noise power induced by RIN in a single PD, the noise power expression for  $I_{sub}$  omits the term  $I^{RIN^2} [1 + 2J_0(\phi_b)]$ . Furthermore, given the inherent weakness of  $P_s$ ,  $P_+$  and  $P_-$ , the coefficients of all terms containing  $J^2(\phi_b)$  experience significant reductions. Simultaneously, the noise power of  $I_{add}$  decreases as the coefficient of the  $J_{2n-1}^2(\phi_b)$  term in the noise power expression diminishes. Theoretical analysis suggests that the new scheme exhibits lower noise power than the conventional scheme in harmonic extraction, implying enhanced sensitivity.

Similarly, the expressions for  $P_{sub}^{dm}$  and  $P_{add}^{dm}$  in the differential-mode measurement are as follows.

$$\begin{aligned} P_{sub}^{dm} &= I^{RIN^2} [J_0^2(\phi_b) (P_s + P_-) P_+ + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) \\ &\quad (P_s + P_-) P_+ + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) (P_s + P_-)]. \end{aligned} \quad (23)$$

$$\begin{aligned} P_{add}^{dm} &= I^{RIN^2} [1 + J_0^2(\phi_b) + 2J_0(\phi_b) \\ &\quad + 4 \sum_{n=1}^{\infty} J_{2n}^2(\phi_b) + 4 \sum_{n=1}^{\infty} J_{2n-1}^2(\phi_b) P_+]. \end{aligned} \quad (24)$$

The suppression of noise powers in  $I_{sub}$  and  $I_{add}$  is evident during the differential-mode measurement. Similarly, the sensitivity is enhanced.

## V. EXPERIMENTAL VERIFICATION

In order to verify the effectiveness of the improved fusion algorithm, a series of verification experiments are carried out. The optical configuration is arranged as depicted in Fig. 1. Two sensors are driven by an ASE light source (BG-ASE-M1-C, Beogold Technology, Xiamen, China) with a central wavelength of 1550 nm and a spectral width of 35 nm. The MIOC (GC15YA3016, Chongqing, China) has a central wavelength

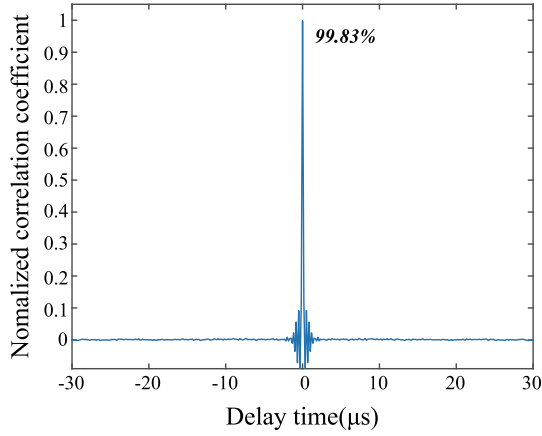


Fig. 3. Normalized correlation coefficient of two IFOGs.

of 1550 nm and a working spectrum width of 50 nm. Each polarization-maintaining fiber coil measures 1 km in length and 0.13 m in diameter. PD A and PD B (LSIPD-UL0.3-B-4JKFC, Minguang Technology, Beijing, China) are calibrated to ensure consistency. The signal acquisition system adopts PXI-5922 (National Instruments, Austin, USA) with a sampling rate of 6 MHz.

The inconsistency in the optical intensity of the two paths affects noise suppression performance, and we employ various methods at both the device and signal processing levels for compensation. At the device level, we opt for a coupler with a coupling ratio close to 1:1 and PDs exhibiting high consistency in responsivity. On the signal processing level, we incorporate a compensation coefficient into the computation. Specifically, before the experiment, the light intensity signals of the two IFOGs are collected and the average value is calculated. The compensation factor is determined by the ratio of their mean values. Compensating can be achieved by multiplying the calibration factor to the signals detected by PD A or PD B. After completing the compensation for average optical intensities, it is necessary to verify the noise correlation between the two sensors. Optical signals in PD A and PD B are recorded without applying any modulation or excitation signals. The calculated normalized correlation coefficient between them is illustrated in Fig. 3. It is evident that the noise correlation between the two IFOGs reached 99.8%. Adequate noise correlation establishes the foundation for noise suppression.

#### A. Measurement Under Excitation by PZTs

To demonstrate the system's ability to extract common-mode and differential-mode signals, we apply signals using two pairs of PZTs to impose phase excitation. The optical fiber is tightly wound around the PZTs, and the arrangement of the two pairs of PZTs is illustrated in Fig. 4. Voltage signals are applied to these PZTs. PZTA1 and PZTB1 are used for differential-mode sinusoidal excitations, generating reverse-phase sinusoidal signals with a frequency of 5 Hz. Similarly, PZTA2 and PZTB2 are employed for common-mode sinusoidal excitations, producing in-phase sinusoidal signals with a frequency of 10 Hz.

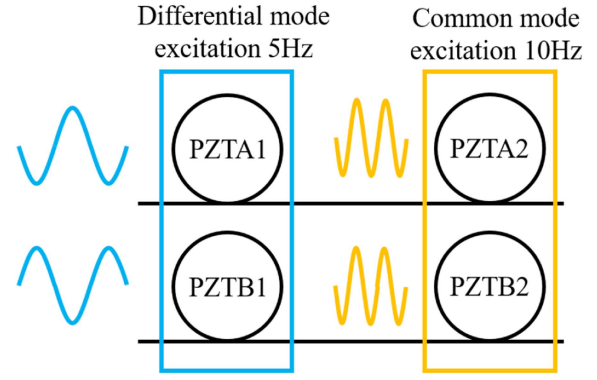


Fig. 4. Schematic diagram of applying common-mode signal and differential-mode signal using PZTs.

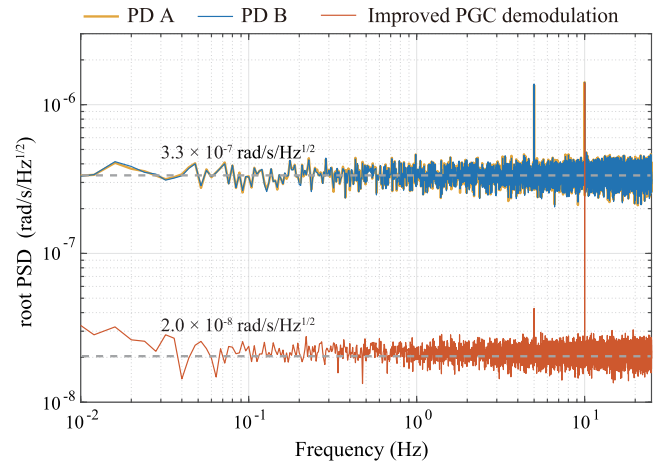


Fig. 5. The PSD of conventional and improved fusion algorithm under common mode. The differential-mode disturbance and intrinsic noise are suppressed.

The common-mode signal extraction experiment is first conducted. The source-sharing configuration is positioned on a horizontal surface. Two reverse-phase modulation signals, with a modulation frequency of 297.9 kHz and controlled modulation depths of 2.7 rad, are applied to the two MIOCs. We respectively calculate the equivalent angular velocity signals in each IFOG using the conventional PGC algorithm and the common-mode signal using the improved fusion algorithm. The root PSD is calculated for all three cases, and the results are depicted in Fig. 5. The differential-mode disturbance and intrinsic noise are suppressed, and the residual 5 Hz disturbance primarily arises from slight inconsistencies between PZTs.

Subsequently, the experiment for extracting the differential-mode signal is performed. The phase difference of the modulation signals between the two MIOCs is set to 0. The modulation frequency remains at 297.9 kHz, and the modulation depths are maintained at 2.7 rad. We respectively calculate the equivalent angular velocity signals in each IFOG using the conventional PGC algorithm and the differential-mode signal using the improved fusion algorithm. The root PSD for all three cases is depicted in Fig. 6. The common-mode disturbance and intrinsic noise are suppressed, and the residual 10 Hz disturbance primarily arises from slight inconsistencies between PZTs.



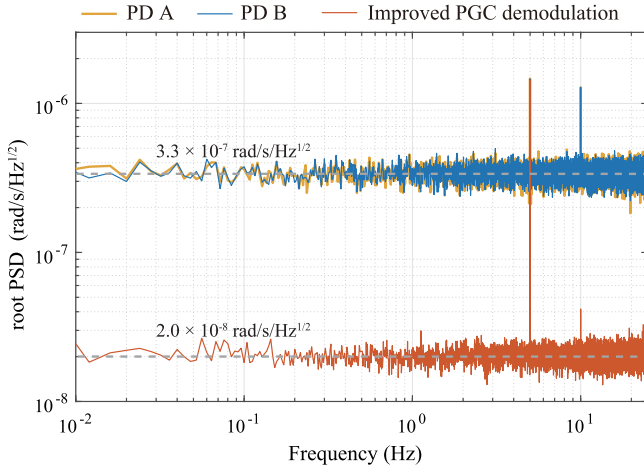


Fig. 6. The PSD of conventional and improved fusion algorithm under differential mode. The common-mode disturbance and intrinsic noise are suppressed.

### B. Measurement With Environmental Disturbances

To further validate the ability of the scheme to remove environmental disturbances, extended observations in complex environments are conducted. The source sharing configuration is placed horizontally on the ground, and the sensitive signal consists of two parts. One component is the projection of the Earth's rotation on the horizontal plane, while the other is the vibrational component from the environment, primarily induced by natural motion and human activities [37].

For common-mode measurements, the phase difference  $\Delta\theta$  between the two IFOGs is set to  $\pi$  rad. The root PSD and Allan standard deviation are illustrated in Fig. 7(a) and (b), respectively. For each single IFOG, the self-noise level in the root PSD is  $3.3 \times 10^{-7} \text{ rad/s/Hz}^{1/2}$ . The ARW and BI in the Allan curve are  $8.0 \times 10^{-4} \text{ deg/h}^{1/2}$  and  $2.0 \times 10^{-3} \text{ deg/h}$ , respectively. Through the improved fusion algorithm, the self-noise level reaches  $2.0 \times 10^{-8} \text{ rad/s/Hz}^{1/2}$ , exhibiting a remarkable reduction of 16.5 fold. The ARW and BI are reduced by 16.0-fold and 10.0-fold, which reach  $5.0 \times 10^{-5} \text{ deg/h}^{1/2}$  and  $2.0 \times 10^{-4} \text{ deg/h}$ , respectively. While suppressing RIN, the vibration signal in the environment remains intact. Benefiting from the noise suppression effect, vibrations generated by human cultural activities (above 10 Hz) are recorded more precisely. The rotation component of the natural seismic signal (1–7 Hz), which is drowned out by self-noise in a single IFOG, is also captured using the new scheme.

For differential-mode measurement, the phase difference  $\Delta\theta$  between the two IFOGs is set to 0. Since the two fiber coils are placed in the same environment and the parameters are identical, the signals sensed by both are highly similar. The differential-mode signal between the both is almost negligible. The root PSD and Allan standard deviation are illustrated in Fig. 8(a) and (b), respectively. The observed signals of each IFOG are consistent with the common-mode measurement experiment. Through the improved fusion algorithm, the self-noise level also reaches  $2.0 \times 10^{-8} \text{ rad/s/Hz}^{1/2}$  under differential-mode measurement. Self-noise still shows a 16.5-fold optimization. The ARW and

BI reach  $5.0 \times 10^{-5} \text{ deg/h}^{1/2}$  and  $2.0 \times 10^{-4} \text{ deg/h}$ , respectively. Compared with a single IFOG, the two properties under differential-mode measurement are also optimized by a factor of 16.0 and 10.0, respectively. Not only the self-noise has been strongly suppressed, but also the common-mode disturbances generated by human cultural activities and natural movements have been almost completely eliminated. The residual response after fusion originates from slight differences in environmental disturbances. Their peak-like nature significantly differs from spectra of signals such as rotational seismic signals and gravity gradient signals. Furthermore, they do not compromise the long-term stability of IFOGs, thereby ensuring their suitability for applications in related fields. The results provides a new prospect for some applications requiring strict control of the environment [23], [38].

According to the experimental results, all the performance parameters are summarized in Table II. The new scheme achieves all-round performance improvement while adding the disturbance removal capability. The performance improvement demonstrated by the new scheme far exceeds the benefits brought about by simply increasing the number of sensors [39] or doubling the coil area [40].

## VI. DISCUSSION

To address the challenges posed by intrinsic noise and eliminate the impact of environmental disturbances on precision measurement, we propose an improved fusion algorithm based on the source-sharing configuration. In this study, the noise mechanism is studied in detail, and further, the contribution of each intrinsic noise under the PGC algorithm is calculated theoretically. The improved fusion algorithm provides a way to extract common-mode signal or differential-mode signal, and the noise power caused by RIN is greatly reduced in the process of harmonic extraction. Different from the optical path reconstruction, the optical structure in this scheme is simple and avoids the non-ideal introduced by optical reference arms. In addition, compared with other algorithms such as artificial neural network, this scheme does not need training data and shows excellent generalization ability in the experiments.

We compare this work with some high-performance IFOGs in industry and academia. The comparison items are listed in Table III. Both SRS-2000 and ASTRIX200 [41] are strategic-grade IFOGs for space navigation. iXblue 2nd mockup [42] is the leading sensor for rotational seismology. Dual light source configuration [39] and Mode-division multiplexing scheme [40] are new achievements in academia. The former uses polarization multiplexing and two independent light sources to improve IFOG performance, and the latter uses mode division multiplexing to double the closed area. Compensation scheme is a noise suppression scheme, which uses the conventional PGC algorithm as the demodulation method [12]. This study achieves self-noise and ARW outperforming all the above schemes with the smallest closed area, and demonstrates the ability of environmental noise suppression. To our knowledge, this is the most comprehensive noise and disturbance suppression scheme in IFOGs.



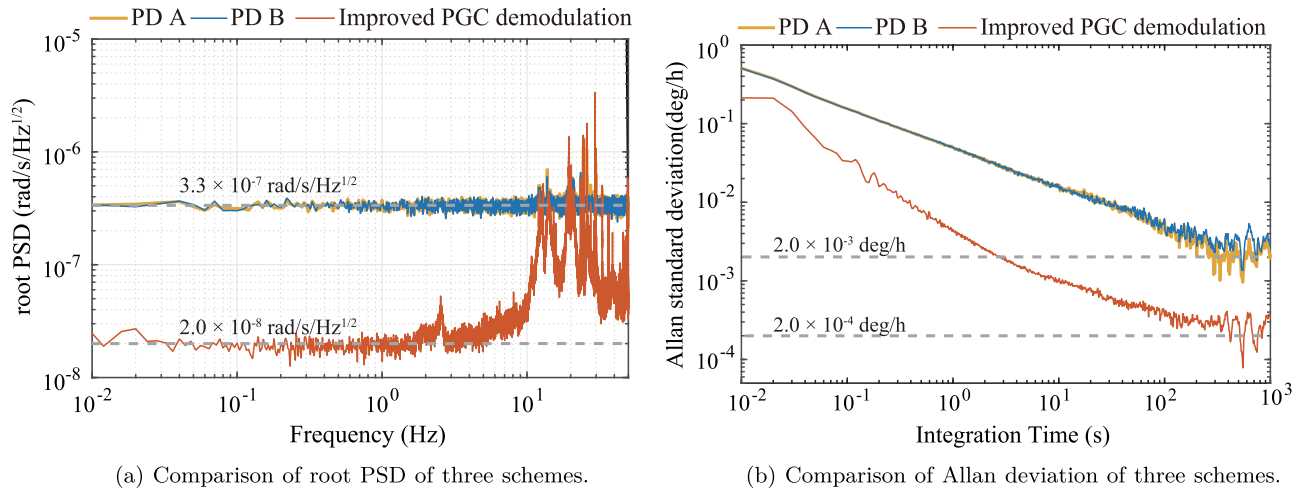


Fig. 7. Performance analysis under common-mode measurement: (a) root power spectral density and (b) Allan standard deviation. Different line widths are used to distinguish.

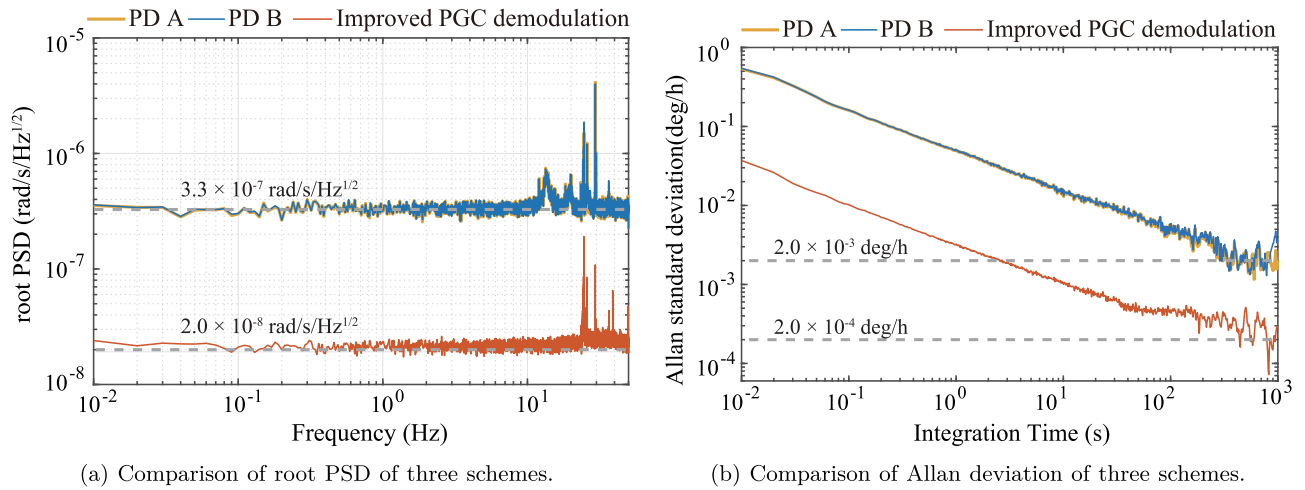


Fig. 8. Performance analysis under differential-mode measurement: (a) root power spectral density and (b) Allan standard deviation. Different line widths are used to distinguish.

TABLE II  
SUMMARY OF THE PERFORMANCE INVOLVED IN THE EXPERIMENTS

| Category              | Self-noise<br>(rad/s/Hz <sup>1/2</sup> ) | Improved<br>factor | ARW<br>(deg/h <sup>1/2</sup> ) | Improved<br>factor | BI<br>(deg/h)        | Improved<br>factor | Disturbance<br>Removal | Coil area<br>(m <sup>2</sup> ) |
|-----------------------|------------------------------------------|--------------------|--------------------------------|--------------------|----------------------|--------------------|------------------------|--------------------------------|
| Minimum configuration | $3.3 \times 10^{-7}$                     | 1                  | $8.0 \times 10^{-4}$           | 1                  | $2.0 \times 10^{-3}$ | 1                  | ✗                      | 32.5                           |
| New scheme            | $2.0 \times 10^{-8}$                     | 16.5               | $5.0 \times 10^{-5}$           | 16.0               | $2.0 \times 10^{-4}$ | 10.0               | ✓                      | $32.5 \times 2$                |

TABLE III  
PERFORMANCE COMPARISON OF THIS WORK AND HIGH-PERFORMANCE IFOGS

| Category                           | Self-noise<br>(rad/s/Hz <sup>1/2</sup> ) | ARW<br>(deg/h <sup>1/2</sup> ) | BI<br>(deg/h)        | Disturbance<br>Removal | Coil area<br>(m <sup>2</sup> ) |
|------------------------------------|------------------------------------------|--------------------------------|----------------------|------------------------|--------------------------------|
| SRS-2000 [42]                      | $7.6 \times 10^{-8}$                     | $2.6 \times 10^{-4}$           | $2.4 \times 10^{-4}$ | ✗                      | 125                            |
| ASTRIX200 [42]                     | $6.1 \times 10^{-7}$                     | $2.1 \times 10^{-4}$           | $5.0 \times 10^{-4}$ | ✗                      | 165                            |
| iXblue 2 <sup>nd</sup> mockup [43] | $4.3 \times 10^{-8}$                     | $1.5 \times 10^{-4}$           | /                    | ✗                      | 90                             |
| Dual-source configuration [40]     | $1.3 \times 10^{-7}$                     | $4.6 \times 10^{-4}$           | $4.6 \times 10^{-4}$ | ✗                      | 67                             |
| Mode-division multiplexing [41]    | $9.7 \times 10^{-8}$                     | $2.3 \times 10^{-4}$           | $8.8 \times 10^{-4}$ | ✗                      | 67                             |
| Compensation scheme [12]           | $2.4 \times 10^{-8}$                     | $6.0 \times 10^{-5}$           | $2.1 \times 10^{-4}$ | ✓                      | 158                            |
| This work                          | $2.0 \times 10^{-8}$                     | $5.0 \times 10^{-5}$           | $2.0 \times 10^{-4}$ | ✓                      | 65                             |

In precision measurement, adopting common-mode and differential-mode measurements are established strategies for improving performance. For general applications such as inertial navigation, rotational seismology, mechanical measurements [43], etc., employing common-mode measurements as presented in this paper can significantly enhance measurement sensitivity and long-term stability, thereby improving the ability to measure weak signals. In the case of gravity gradient measurement [24], [38], [44], two IFOGs with opposite sensitive directions can be fixed on probes with opposite responses to gravity gradients. This not only deals with self-noise but also effectively suppresses environmental disturbances.

Additionally, this scheme can be combined with a symmetric winding method [45]. Temperature induced phase error and sagnac phase have opposite signs in two coils, thus the impact of temperature gradient can be suppressed. Furthermore, the scheme can be integrated with a linearity compensation algorithm [8] to correct linearity errors caused by uneven frequency response of the components.

## VII. CONCLUSION

We propose an improved fusion algorithm combined with a light source sharing configuration, achieving effective and strong intrinsic noise suppression and disturbance removal capabilities. This work reveals the mechanism of intrinsic noise and provides a detailed analysis of the impact of each type of noise under the conventional PGC algorithm. The improved fusion algorithm jointly processes two interferometric signals, efficiently separating common-mode and differential-mode signals through a concise and effective calculation approach. By adjusting the phase difference between the two modulation signals, flexible switching between common-mode and differential-mode measurements is achieved. Importantly, based on the proposed noise theory, the impact of the primary intrinsic noise RIN, on harmonic extraction is significantly suppressed.

Experimental validations, including measurement under excitation by PZTs and measurement with environmental disturbances, support the theoretical analysis. Compared to a single IFOG, the light source sharing structure using the improved fusion algorithm achieves comprehensive performance improvements: a 16.5-fold reduction in self-noise, 16.0-fold reduction in ARW, and 10.0-fold reduction in BI. Additionally, in each measurement mode, the desired components are fully preserved while suppressing the other component.

Compared with the world-leading IFOGs in industry and academia, the proposed scheme shows remarkable sensitivity performance. In our knowledge, this is the most complete noise treatment scheme. The proposed solution demonstrates the potential for IFOGs to exhibit high-precision measurement capabilities in a broader range of scenarios, promising advancements in inertial navigation, geophysical exploration, and related fields.

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